Validating Data for a Linguistic Resource

Error analysis, agreement, and giving back

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JPC3 — Improving the Czech Wordnet

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Overview

- Why be sure about the data
- Error analysis: beyond one number
- Why two annotators
- Cohen's κ, worked through
- Giving fixes back to ARASAAC, Wikidata, …

Roadmap

1. Why be sure?
2. The new bottleneck
3. Building an explainer
4. Error analysis
5. Agreement
6. Cohen's kappa
7. Giving back

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Why be sure?

- A wordnet is a **trusted source**
- Many tools read it; few check it
- An error you add, everyone inherits
  - e.g. a wrong hypernym copied into every OMW language
- Once released, it is copied, and hard to recall
- The Czech wordnet feeds OMW, sense tagging, AAC, …

The cost of one wrong link

- Project 4 links pictographs for **AAC** users
- A wrong synset tag $\rightarrow$ the wrong picture
- A child who cannot speak is handed the wrong word

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Understanding Is the Bottleneck

- The starter code generates candidates in seconds
  - feminine candidates for all 1,919 role-noun synsets at once
- Machines already verify themselves well
- The slow part is **understanding** what came out
- “Cognitive debt”: skip it and you lose the plot
- Generation got cheap; understanding did not (Litt, 2026)

https://www.geoffreylitt.com/2026/07/02/understanding-is-the-new-bottleneck.html

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Staying in the Loop

Three ways to keep understanding ahead of the machine:

- Explain before you trust: write what the data *says*
- Plot it: a chart shows the outliers a table hides
- **Spot-check** a sample before trusting the whole
  - judge 20 by hand before you generate 2,000

Look at Your Data as You Go

- Check the output at every step of the work
- Dump it in a **viewable format**: TOML, CSV, a plot
- A wrong turn caught early is cheap to undo
- A literate program shows its output inline, by habit

"Actually look at your data!"

Explainers as Literate Programming

- Knuth (1984): write code for **humans** to read
- Prose and code woven together, explanation first
- Litt's explainer is the same idea, authorship flipped
- The agent writes the code; you write the understanding
- Your report is the literate layer over the scripts

Revived for LLMs: "Natural Language Outlines" (2024), "Verifiable Literate Programming" for validating generated code (2026)

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An Explainer, From the Code

- Generate an HTML explainer **from** the finished script
- Four parts: background, intuition, a code tour, a quiz
- One self-contained page, one link to share
- The script stays a normal file, with nothing to keep in sync
- Every starter script now has one, on the course site

Build It Interactively

- React to a draft; each reply refines the same page
- The link stays stable, so you share it once
- Keep every version in the repo: `v0` to `v4`
- The actual prompts behind this explainer:
  - `v0`: “build an explainer for this script”
  - `v1`: “where do the ILI mappings come from? add a SPARQL appendix”
  - `v2`: “add a glossary”
  - `v3`: “add caveats: it is AI-generated and can be wrong”
  - `v4`: “hyperlink the key terms; stamp the provenance”

Make It Repeatable

The same **recipe** works for any project script:

- Point at a script: ask for background, intuition, code, quiz
- Read it, then ask for the next fix
- Keep each version; link the latest

Packaged as the `code-explainer` skill in the repo

Guarding an Explainer

It is AI-generated, so treat it like any other data:

- A visible caveat on the page: it can be wrong, so check the code
- `explainer_check.py`: identifiers, line refs, and links against the source
  - on ours it found nothing wrong, and one unreachable link
- A **provenance** stamp: date, model, source commit
- A second reader, human or model, checks the prose

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Error Analysis

- Take a **random** sample (aim for 200)
- Judge each by hand: right or wrong?
- Precision = correct / judged
- But the single number is the least of it
- Group the mistakes into **types**: that is the payoff
- One real example per type, with a count

Running example: přechylování

Candidate: add a feminine form to a role-noun synset.

(1) Czech [ces]
učitel → *učitel-ka*
teacher.M teacher-F
“(male) teacher → (female) teacher”

- 1,919 role-noun synsets to work from (`noun.person`)
- Easy to generate; the question is which are **right**

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An Error Taxonomy

A sample of 200 generated feminine candidates:

Error type	Count	Example
Correct	171	*lékař* → *lékařka*
Coincidental suffix	12	*past* (not a base + *-ka*)
Wrong sense	9	feminine of a non-personal reading
Already present	8	lemma was in the synset

Precision 0.86 — but the four rows tell the next team where to look

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Why Two Annotators?

- One judge can be **consistently** wrong
  - e.g. always accepting a coincidental *-ka* (*past*→ *pastka*)
- A second judge tests whether the scheme is learnable
- Agreement ≈ reliability of the guidelines
- You work in pairs, so measure it
- Report how you resolved the disagreements

Raw Agreement Overcounts

- “We agreed 85% of the time” sounds good
- But some agreement happens by **chance**
- Two coin-flippers agree half the time
- If almost everything is “correct”, agreeing is easy
- Correct for chance $\rightarrow$ Cohen’s κ

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Cohen's Kappa

Agreement above what chance would give:

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

- p_o: observed agreement (the proportion you both called the same)
- p_e: expected agreement **by chance**, from each annotator's rates
  - if you both say "good" 90% of the time, chance agreement is high
- $\kappa = 1$ perfect; $\kappa = 0$ no better than chance

A Worked Example

200 candidates, two annotators, good vs. bad:

	B: good	B: bad	total
A: good	130	20	150
A: bad	10	40	50
total	140	60	200

- $p_o = (130 + 40)/200 = 0.85$
- $p_e = 0.75 \times 0.70 + 0.25 \times 0.30 = 0.60$

$$\kappa = \frac{0.85 - 0.60}{1 - 0.60} = \mathbf{0.63}$$

Reading Kappa

- Rough scale (Landis & Koch, 1977): > 0.8 very good, 0.6–0.8 substantial
- Our 0.63 is **substantial**: the guidelines mostly work
- Low κ? Fix the guidelines, then re-annotate
- Beware: few categories or a skewed split inflate it
- Report κ together with the raw table

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Why Open? — Not Just the Warm Inner Glow

- Our goal: a better Czech wordnet, *and* a **contribution**
- Open data others can build on
- Your fixes reach every downstream user
- A result that outlives the course

Feedback Upstream

Every error you find is a **gift** to the source.

- ARASAAC: report wrong synset tags; send Czech keywords back
- Wikidata: add or fix the ILI (P5063) and WordNet id (P8814)
- Global Symbols / ARAWN: contribute the matches you validated
- CILI: report synsets with no ILI; DeriNet: report gaps
- PDT: the unmapped residue is data for the maintainers

Give Back: ARASAAC

Project 4 — pictographs for AAC.

- Found a **wrong** synset tag? Report it to ARASAAC
- Missing Czech keyword? Submit the translation
- Note colour and gender variants you matched
  - e.g. the generated xkcd swatch for *chestnut*
- Their pictograms carry synsets, so a fix helps every language

Give Back: Wikidata

Project 5 — Wikidata → ILI → Czech.

- Only a WordNet id (P8814)? Add the **ILI** (P5063)
- Linked to the wrong sense? Correct the statement
- Confirmed a Czech label? Add it, with a reference
  - e.g. add *Nizozemsko* to the *Holandsko* synset
- Cite the wordnet as the source: statements need one

Give Back: PDT and CILI

Project 3 — the sense-tag residue.

- The **unmapped** tags are a bug list for the maintainers
- Split or merged senses? Propose the current synset
- Share your 1.5 → current mapping table
- New concept with no ILI? Ask CILI to mint one

Give Back: DeriNet

Project 2 — derivational relations.

- Validated feminine-motion pairs? Offer them back
  - e.g. *učitel*→ *učitelka*
- A family the wordnet **lacks**? Flag the gap
- A DeriNet error you caught? Report it
- Both resources improve from the same checking

The Virtuous Cycle

- Open data gets better only when fixes **flow back**
- A pull request or an issue beats a private fix
- Cite what you used; credit who made it
- Release under an open licence, loadable with `wn`
- Next year's class starts from your cleaner data

Every new wordnet makes the network richer

Conclusion

- Extraction is scaffolded: **validation** is the work
- Sample, judge, and report precision
- Categorise the errors; the taxonomy guides the next step
- Two annotators, and κ with the table
- Push every fix back upstream

Thanks

- Validate carefully: others will trust your data
- Actually look at your data!
- Code and template: the JPC3 repository

Questions welcome — and so are pull requests

References I

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